

Replacement memo: corrected heavy-tail Section 5.2

SVP simulation audit

15 September 2026

Decision

Section 5.2 is replaced by the Student- $t(2)$ stress study described below. The original `SVP_NEW.pdf` is unchanged because no editable source was available. The previous robust outputs are preserved in `simulations/other_simus/power_robust_legacy_20260915/`.

Calibration

RFPOP uses the biweight loss (called `Outlier` by `robseg`) with ℓ -threshold 3 and $\lambda = c_{\text{RF}} \log n$. The smallest candidate with corrected null F1 at least 0.99 is selected. Median-Mood and Wilcoxon SVP thresholds are then selected to match the RFPOP null F1 target. Calibration and validation use independent $t(2)$ streams with 20,000 and 10,000 series, respectively; Wilson intervals and deterministic seeds are reported in the CSV table below.

Table 1: Validation of selected robust null thresholds.						
Method	Candidate	F1	Lower	Upper	RFPOP target	Seed
RFPOP	3.0000	0.9958	0.9943	0.9969	0.9961	920000.0000
Wilcoxon	1.7500	0.9951	0.9935	0.9963	0.9961	920000.0000
MedianMood	0.0010	0.9979	0.9968	0.9986	0.9961	920000.0000

The corrected Gaussian-PELT null F1 is the external benchmark. Gaussian PELT under $t(2)$ is retained as a stress baseline and is not used as a calibration target because its Gaussian SSE fit produces almost universal false positives.

Before/after metrics

The detailed paired table is indexed by pattern and jump in `paired_comparison.csv`; the compact table below averages changes over the 40 jump sizes within each pattern. Positive changes improve F1, precision, recall and correct-number probability; negative changes improve MSE and segment count.

Replacement Section 5.2

We assess robustness to heavy-tailed noise using independent Student- $t(2)$ errors generated by `ts_generator(type = "student", df = 2, scale = 1)`. The design has $n = 1000$, four patterns

Table 2: Mean new-minus-legacy metric change by pattern and method.

Pattern	Method	F1	Precision	Recall	Correct	MSE	Segments
none	MedianMood	0.0041	0.0069	-0.0015	0.0153	-0.0006	-0.0155
none	PELT	-0.0258	-0.0131	-1.0000	0.0000	0.0000	0.0000
none	RFPOP	0.0476	0.0666	-0.0050	0.1105	-0.0068	-0.2020
none	Wilcoxon	0.0031	0.0066	-0.0040	0.0173	-0.0014	-0.0173
rand1	MedianMood	-0.0418	-0.1757	-0.0019	0.2297	-0.0895	0.8985
rand1	PELT	-0.0202	-0.0117	-0.0421	0.0000	0.0000	0.0000
rand1	RFPOP	-0.0690	-0.0968	-0.0686	-0.0050	0.0008	-0.2902
rand1	Wilcoxon	-0.0537	-0.0510	-0.0565	-0.0365	-0.0100	-0.1928
up	MedianMood	-0.0397	-0.0910	-0.0215	0.1145	-0.1017	0.4695
up	PELT	-0.0202	-0.0117	-0.0417	0.0000	0.0000	0.0000
up	RFPOP	-0.1183	-0.1049	-0.1052	0.0000	0.0734	-0.2760
up	Wilcoxon	-0.0620	-0.0634	-0.0609	-0.0285	-0.0261	-0.0975
updown	MedianMood	-0.0376	-0.2366	-0.0057	0.1350	-0.0609	0.8145
updown	PELT	-0.0202	-0.0117	-0.0413	0.0000	0.0000	0.0000
updown	RFPOP	-0.0858	-0.1698	-0.0746	-0.0400	0.0060	-0.6450
updown	Wilcoxon	-0.0633	-0.0647	-0.0622	-0.0312	0.0011	-0.1077

(none, up, updown and rand1), jump sizes from 0.1 to 4 in increments of 0.1, and 100 replications. The fixed Gaussian PELT fit with penalty $2 \log n$ is retained as an intentionally misspecified stress baseline. The robust competitors are null-calibrated RFPOP with biweight loss and ℓ -threshold 3, Median-Mood SVP, and Wilcoxon SVP.

For every method, the mandatory terminal endpoint n is removed before one-to-one matching within the framework tolerance. We report precision, recall, F1, correct-number probability, signal MSE, number of fitted segments, and localization error. RFPOP is calibrated to at least 0.99 corrected null F1. The Median-Mood and Wilcoxon thresholds are selected to match that target and all SVP calls use the `right` pruning mode. Their length-dependent scaling uses K , passed in the fitting code as `oracle_segments`, the known number of data-generating segments in this simulation. This oracle is a design device and is not assumed available in practical data analysis.

$$\gamma_W(c, K) = c \sqrt{\frac{(n/K)^3}{12}}, \quad \gamma_M(\alpha, K) = \chi_{1, 1-\alpha_s}^2, \quad \alpha_s = 1 - (1 - \alpha)^{1/(n/K-1)}.$$